$\boxed{\text{Author:}}$ Unfortunately, the concept of a function as a formula relating x and y has long been rooted in the minds of students. This is, of course, quite wrong. A function and its formula are very different entities. It is one thing to define a function as a mapping of one set (in our case it is a numerical set) onto another, in other words, as a "black box" that generates a number at the output in response to a number at the input. It is quite another thing to have just a formula, which represents only one of the ways of defining a function. It is wrong to $\boxed{\text{identify}}$ a function with ~~its~~ a formula giving its analytical description (unfortunately it happens sometimes).

$\boxed{\text{Reader:}}$ It seems that after the discussion in the previous dialogue about the function, such identification in a general case is automatically invalidated. However, if we confine ourselves only to $\boxed{\text{numerical functions}}$ and if we bear in mind that working with a function we always use a formula to describe it, a question arises: Why is it erroneous to identify these two notions? Why should we always emphasize the difference between the function and its formula?

$\boxed{\text{Author:}}$ I will tell you why. First, not every formula defines a function. Actually, at the end of the previous dialogue we already had such an example. I shall give you some more: $y=\frac{1}{\sqrt{x}}+\frac{1}{\sqrt{-x}}$, $y=\log(x)+\log(-x)$, $y=\sqrt{\sin x-2}$, $y=\log(\sin x-2)$, etc. These formulas do not represent any functions.

Second (and this is more important), not all functions can be written as formulas. One example is the so-called $\boxed{\text{Dirichlet function}}$ which is defined on the real line:

$$y=\begin{cases}1 & \text{if } x \text{ is a rational number}\\ 0 & \text{if } x \text{ is an irrational number}\end{cases}$$

$\boxed{\text{Reader:}}$ You call this a function?

$\boxed{\text{Author:}}$ It is certainly an unusual function, but still a function. It is a mapping of a set of rational numbers to unity and a set of irrational numbers to zero. The fact that you cannot suggest any analytical expression for this function is of no consequence (unless you invent a special symbol for the purpose and look at it as a formula).